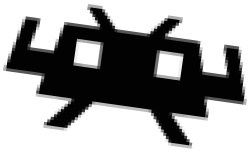
RetroArch



*My harvesting and personal contributions of digital media and content in support of retro gaming and computing from **1976***

Since November 2021, select digital content pulled off of my Drive folder to make this into a plug & play **RetroArch** configuration that loads from a Debian 13 **Trixie** boot image for both a user & developer friendly **Raspberry Pi 4** or **5** into a Wayland desktop (or the Linux console).

```
$ lsblk -S  # find your target device NAME (SD card, USB stick)
$ xzcat RA-playlists-rpi-distro.img.xz | sudo dd of=/dev/[NAME] bs=4k status=progress
$ sync
```

The first boot may take a little longer ~ *be patient*. Once that has settled, use the system panel controls to configure your **`network`**, **`sound`** (*might need to right-click* Volume *control to select an output profile*), and turn on **`bluetooth`** and pair any devices.

📝 Notes

- ➔ do read the included **`HELP`** document off the Desktop or the console Playlists
- ➔ both desktop and console auto login as user **`pi`** with password **`Freedom`**
- ➔ console menu: press left `Alt` for a prompt to switch between the running shell profiles
- ➔ automatic emulation tweaks are made from Pi 4/400 or 5/500 model detection
- ➔ enhanced integration for Sony **PS5 DualSense** controllers and **Pi 500+** RGB keyboard
- ➔ **`130gb`** for systems emulation and content with **`36gb`** of multimedia, overlays, and themes

⏳ Latest Changes

Using the console Playlists menu **`upgrade`** option retrieves the latest changes listed here. Changes to content and themes are not part of the upgrade process - that is best done off the 💾current image.

27-Oct-2026

- latest working git commits off EmulationStation and RetroArch v1.22.2 release
  - refreshed core info and rebuilt cores: blueMSX, gbSP, Genesis Plus GX, Stella, Opera, PicoDrive, PUAE, PCSX ReARMed, PPSSPP, with Mednafen for Lynx, Neo Geo Pocket + Color, and Virtual Boy
  - make-retroarch.sh enhancements; slimmed down libretro-super batch build to relevant cores used here, but without the long builds with FBneo, MAME 2003-plus and MAME
  - updated COIN OP theme Zmax with a different scene selection between startups
- improve curated ES playlists: its organization with added content and enabled achievements
- major refactoring for playlists by consolidating its multimedia into pi's home Media folder

⌨️ Technical

Layout

This **RetroArch** configuration is designed to be ‘plug & play’. And while it allows for flexibility to make some runtime modifications, it is designed to revert back to its original state after reboot. This is intentional, not to annoy you, but to protect from changes that lead to incidental havoc to gameplay.

That said, feel free to drop to a terminal shell prompt by using left `Alt` then `2` off the console menu to check out the differences between a default and its startup template configuration:

```
$ ll *.cfg
$ diff default.cfg template.cfg | less
$ micro template.cfg
```

Thus, **`template.cfg`** gets copied over to **`retroarch.cfg`** ⟶ which points to the pi temporary runtime memory space. It is the “base” configuration that gets loaded first. Subsequently, other configuration files listed may be “appended” in an order received to override base values, as concise and appropriate as needed to run the content.

Overclocking

These are safe values that bump the CPU (+ GPU) clocks up in `/boot/firmware/config.txt`

```
[pi4]
# enables highest gpu settings at the expense of dual displays
hdmi_enable_4kp60=1
arm_freq=1980
#gpu_freq=550
#over_voltage=4

[pi5]
dtparam=nvme
dtparam=pciex1_gen=3
#kernel=kernel8.img
arm_freq=2760
gpu_freq=920
#over_voltage=3
```

If using an NVME drive on Pi 5, enable it to boot:

```
$ sudo rpi-eeprom-config --edit

[all]
BOOT_UART=0
POWER_OFF_ON_HALT=1
BOOT_ORDER=0xf461
PCIE_PROBE=1
WAKE_ON_GPIO=0
```

Startup

As noted, the **`pi`** user account is configured to auto login in both console and desktop modes. Its shell profile **`.bashrc`** script initiates Pi **`MODEL`** detection and prepares the console playlists menu. There is also an automated startup task script **`autostart.sh`** between modes to assure the machine's runtime environment is prepared for optimized use.

Cores

There is a convenience script that can fetch and build RetroArch and some key emulation cores:

```
$ make-retroarch.sh
```

This table represents the active cores installed and referenced within the various playlists. It is useful to know the name of the core for **`Netplay`** connections to another host – as both the core and the content should be (optimally) identical:

The 3DO Company (Opera)	Nintendo - Game Boy Advance (gpSP)
Arcade (FinalBurn Neo Light)	Nintendo - Game Boy / Color (Gambatte)
Arcade (MAME)	Nintendo - GameCube (Dolphin)
Arcade (MAME 2003-Plus)	Nintendo - NES / Famicom (Mesen)
Apple II / IIGS (MAME)	Nintendo - 64 (Mupen64Plus-Next)
Apple Macintosh / Color Classic (MAME)	Nintendo - Pokémon Mini (PokeMini)
Atari - 2600 (Stella)	Nintendo - SNES / SFC (snes9x or bsnes-jg)
Atari - 7800 (ProSystem)	Nintendo - Virtual Boy (Beetle VB)
Atari - 800 (Atari800)	Sega - Dreamcast / Naomi (Flycast)
Atari - Lynx (Beetle Handy)	Sega - MS/GG/MD/CD (Genesis Plus GX)
Commodore - Amiga 500 / 1200 / CD32 (PUAE)	Sega - MS/MD/CD/32X (PicoDrive)
Commodore - C64 (VICE x64sc, accurate)	Sega - Saturn (Yabause)
Commodore - PET (VICE xpet)	Sharp X68000 Compact (PX68k)
Commodore - Plus/4 (VICE xplus4)	SNK - Neo Geo CD (NeoCD)
Commodore - VIC-20 (VICE xvic)	SNK - Neo Geo Pocket / Color (Beetle NeoPop)
DOS (DOSBox-Pure)	Sony - PlayStation (PCSX ReARMed)
GCE - Vectrex (vecx)	Sony - PlayStation Portable (PPSSPP)
Mattel - Intellivision (FreeIntv)	Sony - PlayStation 2 (LRPS2) *tbd for aarch64
MSX/SVI/ColecoVision/SG-1000 (blueMSX)	Tandy - Color Computer (MAME)
NEC - PC Engine / CD (Mednafen_PCE)	Tandy - TRS-80 Model I (MAME)
Nintendo - 3DS (Citra) *tbd for aarch64	Texas Instruments - TI-99/4A (MAME)

Actively used and participated in MAME, VICE, and Stella developments as standalone emulators since 1999 on Linux and Windows. The suite of UAE emulators would follow. Then RetroArch replaced them all in 2021.

🕹️ Emulation Station

In addition to my custom themes for Playlists to launch an **exhaustive** list of content out of `RetroArch`, a comprehensive list of **fun games** using `Emulation Station` is added for its multiple gaming themes with a modern kiosk look & feel experience.

Collections	*All Games*	*Favorites*	*Never Played*
Last Played	*Action*	*Beat'em Up*	*Educational*
Fight	*Pinball*	*Platform*	*Race*
Role-Playing	*Shoot'em Up*	*Shooter*	*Sports*

Arcade			**1976**
10-yard Fight	Gladiator	NBA Hangtime	Space Duel
1994: The Loop Master	Gorf	Pac-Man	Space Invaders
2 on 2 Open Ice Challenge	Gravitar	Phoenix	Spiders
Asteroids	Gun.Smoke	Pigskin 621ad	Stone Ball
Berzerk	Gyruss	Popeye	Street Fighter II: CE
Black Tiger	Joust	Pop'n Pop	Sunset Riders (4 Players)
Blasteroids	Jungle King	Q*bert	Super Punch-out!!
Bubble Bobble	Karate Champ	Rampage	Tapper
Burger Time	Kung-fu Master	Rampart	Tecmo Bowl
Carnival	Lady Bug	Robotron: 2084	Time Pilot
Cyberball	Life Force	Rolling Thunder	Time Pilot '84
Defender	Make Trax	Rolling Thunder 2	Trog
Donkey Kong	Mappy	R-Type	Ultimate Tennis
Elevator Action	Martial Masters	R-Type II	Vampire: The Night Warriors
Frogger	Mercs	Rush'n Attack	Vanguard
Galaga	Moon Patrol	Salamander 2	Wizard Of Wor
Galaxian	Mortal Kombat 2	Satan's Hollow	Xevious
Gauntlet 2	Mr. Do!	Scramble	X-Men (4 Players)
Ghosts'n Goblins	Mr. Do's Castle	Sea Wolf	Zaxxon
Ghouls'n Ghosts	Ms. Pac-Man	Sea Wolf II	

Apple 2			**1977**
Beyond Castle Wolfnstn	Game of the States	Ultima I	United States Adventure
Beyond Pinball	Injured Engine	Ultima II: Killer Char	Yahtzee
Castle Wolfenstein		Ultima III: Exodus	

Atari 2600			1977
Air-Sea Battle	Demon Attack	Pac-Man	Stargate
Asteroids	Dodge 'Em	Phoenix	Star Voyager
Atlantis	Dragonfire	Pitfall II: Lost Caverns	Super Challenge Baseball
Basketball	Enduro	Pitfall!	Super Challenge Football
Battlezone	Frogs and Flies	RealSports Boxing	Threshold
Berzerk	Ice Hockey	RealSports Football	Title Match Pro Wrestling
Bowling	Krull	RealSports Tennis	Tunnel Runner
Boxing	Lock 'n' Chase	River Raid	Vanguard
Breakout	Medieval Mayhem	Robot Tank	Video Pinball : Arcade Pinball
Carnival	MegaMania	Skiing	Warlords
Chopper Command	Missile Command	Space Invaders	Yars' Revenge
Combat	Ms. Pac-man	Space War	
Cosmic Ark	Night Driver	Spider Fighter	

Intellivision			1979
Advanced D&D	Lock 'N Chase	Sea Battle	Thunder Castle
Astrosmash	Major League Baseball	Snafu	Venture
BurgerTime	Pinball		

Commodore VIC 20			1980
Amok	Kids On Keys	Raid on Fort Knox	Star Trek - S.O.S
Avenger	Mega-Cart (2009)	Sargon II Chess	Sub Chase
Dragonfire	Number Crunch	SeaWolf	Super Amok
Fun with Music	Number Nabber	Serpentine	Super Slot
Garden Wars	Omega Race	Shamus	Sword of Fargoal
Gorf	Pipes	Speed & Bingo Math	Type Attack
Home Babysitter	Quikman (4kb ROM)	Spiders of Mars	Visible Solar System
Jupiter Lander	Radar Rat Race	Spills and Fills	

Commodore 64			1982
Advanced Pinball Simulator	Dropzone	Karateka	Stormlord
Afterburner	Druid	Kung-Fu Master	Stunt Car Racer
Arkanoid	First Samurai	Match Day II	Super Robin Hood
Battle Valley	Flimbo's Quest	Metro-Cross	Thrust
Beach Head	Ghostbusters	MicroProse Soccer	Toy Bizarre
Beach Head II	Ghosts'n Goblins	Mikie	Track & Field
Boulder Dash	Grand Prix Circuit	Miner 2049er	Turbo Outrun
Bruce Lee	Gunship	Pac-Mania	Way of the Exploding Fist
Bubble Bobble	Hardball	Platoon	Winter Games
California Games	H.E.R.O.	Raid on Bungeling Bay	Wizball
Caveman Ugh-lympics	Hyper Sports	Raid over Moscow	Wolfling
Commando	Impossible Mission II	Rescuing Orc	World Class Leaderboard
Decathlon	Infiltrator	Rolling Thunder	World Games
Delta	Into the Eagle's Nest	Sanxion	World Karate Championship
Digiloi	Jumpman	Scooby and Scrappy-doo	X-Out

ColecoVision			1982
Bump 'n' Jump	DragonFire	Ken Uston	Qbert's QUBES
Burgertime	Frenzy	Blackjack-Poker	Space Fury
Carnival	Frogger	Keystone Kapers	Spy Hunter
Centipede	Frogger II	Lady Bug	Super Cobra
Cosmic Avenger	Galaxian	Montezuma's Revenge	Time Pilot
Decathlon	Gateway to Apshai	Mr. Do!	Tutankham
Defender	Jumpman Junior	Mr. Do!'s Castle	Zaxxon
Donkey Kong	Jungle Hunt	Popeye	

Vectrex			1982
Armor Attack	Pole Position	Space Wars	Star Castle
Berzerk	Rip-Off	Spin Ball	Star Trek
MineStorm	Scramble		

Daphne			1983
Astron Belt	Dragon's Lair	Space Ace	Super Don Quix-Ote
Cliff Hanger	Drgn's Lair II: Time Warp		

Nintendo Entertainment System			1985
Adventure Island II	Double Dragon II	Little Samson	Super Mario Bros.
Baseball Stars II	Excitebike	Low Gravity Man	Super Mario Bros. 2
Batman	Final Fantasy	Mega Man 2	Super Mario Bros. 3
Battletoads	Gargoyle's Quest II	Mega Man 3	Super Turrican
Bionic Commando	Garry Kitchen's	Mega Man 5	Tecmo Bowl
Blades of Steel	BattleTank	Metroid	Tecmo Cup
Castlevania	Gradius	Mike Tyson's Punch-Out!!	Tecmo NBA Basketball
Castlevania II	Gun Nac	Ninja Gaiden	Tecmo Super Bowl
Castlevania III	Hammerin' Harry	Ninja Gaiden II	Tecmo World Cup
Contra	IronSword	Power Blade 2	TMNT III
Crystalis	Iron Tank	R.B.I. Baseball 3	Top Players' Tennis
Cyberball	Kabuki Quantum	River City Ransom	Tetris
Cyber Stadium Series	Fighter	Rollerball	Ufouria
Disney's DuckTales	Kid Icarus	Shockwave	Wizards & Warriors III
Donkey Kong	Kirby's Adventure	Solstice	Zelda II: Adv of Link
	Legend of Zelda	Spot	

Sega Master System			1985
Alien Syndrome	Ghouls'n Ghosts	Operation Wolf	Space Harrier
Astérix	Golvellius	Ottifants	Spider-Man vs. The
Bubble Bobble	Joe Montana Football	Phantasy Star	Kingpin
Castle of Illusion	Jurassic Park	Prince of Persia	Submarine Attack
Choplifter!	Kenseiden	Rambo II	The Simpsons
Desert Strike	Laser Ghost	Reggie Jackson Baseball	Ultima IV
Double Dragon	Mercs	R-Type	Wonder Boy III
Fantasy Zone II	Ninja Gaiden		World Soccer

Atari 7800			1986
Alien Brigade	Dig Dug	Kung-Fu Master	Pete Rose Baseball
Asteroids	Donkey Kong	Mario Bros.	Pole Position II
Centipede	Double Dragon	Mean 18 Ultimate Golf	Robotron: 2084
Choplifter!	Galaga	Midnight Mutants	Tomcat F-14
Commando	Ikari Warriors	Ms. Pac-Man	Touchdown Football
Crossbow	Joust	Ninja Golf	Xevious
Dark Chambers	Karateka	One-on-One Basketball	

Commodore Amiga			1987
Agony	Dune II	Paperboy	Stardust
Aladdin	Enchanted Land	Phantasie III	Supaplex
Alien Breed II	Exile	Pinball Dreams	Superfrog
Altered Beast	Fire & Ice	Pinball Fantasies	Super Stardust
Apidya	Garrison	Pinball Illusions	SWIV
Banshee	Garrison II	Pinball Wizard	The Ball Game
Benefactor	Giganoid	Project-X	The Immortal
Brides of Dracula	Goal!	QIX	The King of Chicago
Cannon Fodder	Gods	Qwak	The Secret of Monkey Island
Castle Kingdoms	Great Courts 2	Rampart	Transarctica
City Defence	Hostage : Rescue Mission	Rick Dangerous	Traps 'n Treasures
Civilization	It Came from the Desert	Roadkill	TV Sports Basketball
Colonization	Lemmings	Rubicon	Uridium 2
Commando	Lemmings 2 : The Tribes	Ruffian	Virocop
Cyberpunks	Lionheart	Ruff 'n' Tumble	Warlock
Datastorm	MegaLoMania	Sensible World of Soccer	Wings
Defender of the Crown	Monkey Island 2	Shadow of the Beast	Wings of Fury
Deluxe Galaga	Oil Imperium	Shanghai	Zany Golf
Deluxe Pac-man	Operation Thunderbolt	Slam Tilt	Zool 2
Desert Strike	Panza Kick Boxing	Speedball 2	
Double Dragon III			

TurboGrafX-16			1987
Bloody Wolf	Neutopia	The Legendary Axe	TV Sports Basketball
Magical Chase	Neutopia II	The Legendary Axe II	TV Sports Football
		Time Cruise	TV Sports Hockey

Sega Genesis			1988
Air Buster	Ghostbusters	Micro Machines 2	Sonic The Hedgehog 2
Aladdin	Golden Axe	Micro Machines: Military	Steel Empire
Alien Soldier	Golden Axe II	Mic Machines: Turbo 96	Street Fighter II
Castlevania	Holyfield's Boxing	Mortal Kombat III	Sub-Terrania
Dinosaurs for Hire	Jerry 's Pigskin Footbrawl	MUSHA	Super Street Fighter II
Dragon's Fury	John Madden Football	Mutant League Football	Team USA Basketball
Dune	'93	Mutant League Hockey	Toy Story
Faery Tale Adventure	King's Bounty	NBA Live 98	Two Tribes: Populous II
Exodus	Landstalker	NHLPA Hockey 93	Vectorman 2
Exo Squad	Lightening Force	Pagemaster	Virtual Pinball
Fatal Fury 2	Lotus II	Pat Riley Basketball	Wardner
Formula One	Man Overboard!	Pete Sampras Tennis	WeaponLord
Frank Thomas Baseball	Master of Monsters	PGA Tour 96	World Trophy Soccer
Gain Ground	Mazin Saga	Phantasy Star IV	X-Men 2: Clone Wars
Gargoyles	Mega Bomberman	Shinobi III	Ys III
Gauntlet IV	Mega Man	Skitchin'	

Super Nintendo Entertainment System			1990
Aero Fighters	Kirby Super Star	PGA Tour Golf	Super Mario Kart
Animaniacs	Legend of Zelda	Phalanx	Super Mario RPG
Battletoads -	Lemmings 2	Pop'n TwinBee	Super Mario World 2
Battlemaniacs	Mega Man X	Pacific Theater	Super Metroid
Chrono Trigger	Mega Man X2	Operations	Super Punch-Out!!
Donkey Kong Country 2	Mega Man X3	Q*bert 3	Super Smash T.V.
Donkey Kong Country 3	Micro Machines	R-Type III	Tecmo Super Baseball
Extra Innings	Micro Machines 2	Space Megaforce	Tecmo Super Bowl
F1 Pole Position 2	NBA Jam Tournament	Star Fox	Tecmo Super Bowl II
Final Fantasy VI	NBA Showdown	Stone Protectors	Tecmo Super Bowl III
Firepower 2000	NHL '94	Street Fighter II Turbo	Tecmo Super NBA
Hagane	Operation Logic Bomb	Super Baseball 2020	Tiny Toon Adventures
Illusion of Gaia	Peace Keepers	Super Dropzone	

Sega CD			1991
Ecco: The Tides of Time	Popful Mail	Sonic the Hedgehog CD	Soul Star
Eternal Champions			

Sony PlayStation			1994
Asteroids	Darkstone	Mega Man Legends 2	Star Wars: Dark Forces
Big Strike Bowling	Diablo	Megaman: Battle+Chase	Star Wars: Masters
Bishi Bashi Special	Hot Shots Golf 2	Missile Command	Star Wars: Reb Assault II
Blood Omen	Madden NFL 2001	NBA Hoopz	Tiger Woods PGA Tour
Bowling	Medal of Honor	NFL GameDay	Twisted Metal 2
Castlevania: Symphony	MediEvil	Pong: The Next Level	WarCraft II: Dark Saga
Championship Motocross	MediEvil 2	Rage Racer	Xena: Warrior Princess
Crash Bandicoot: Warped	Mega Man 8	Resident Evil 3: Nemesis	Xevious 3D-G+

Sega 32X			1994
Chaotix Kolibri	Mortal Kombat II NBA Jam: Tournament	NFL Quarterback Club Shadow Squadron	Star Wars Arcade Virtua Racing Deluxe

Sega Saturn			1995
Albert Odyssey Burning Rangers Fighters Megamix	Galactic Attack Guardian Heroes Mass Destruction	Panzer Dragoon Panzer Dragoon II Powerslave	Saturn Bomberman Shining Force III Three Dirty Dwarves

Nintendo 64			1996
Beetle Adventure Racing! F-Zero X Legend of Zelda: Majora Legend of Zelda: Ocarina	Mario Golf Mario Kart 64 Mario Party 2 Mario Tennis	Mickey's Speedway USA Ogre Battle: Lrdly Caliber Pokemon Stadium 2 Snowboard Kids	Star Fox 64 Star Wars: Rogue Squad Turok 2: Seeds Of Evil Wave Race 64

Sega Dreamcast			1998
F355 Challenge Gauntlet Legends Hoyle Casino Ikaruga Legacy of Kain: Soul Looney Tunes: Race Magforce Racing Marvel vs. Capcom: Clash Marvel vs. Capcom 2 Maximum Pool Max Steel: Covert	Mortal Kombat Gold Ms. Pac-Man Maze Madness NBA 2K2 NBA Hoopz NCAA College Football 2K2 NFL 2K1 NFL 2K2 NHL 2K2	Phantasy Star Power Stone 2 Pro Pinball Trilogy Rayman 2: Great Escape Record of Lodoss War Red Dog: Superior Power Resident Evil: Veronica SnoCross Championship Sonic Adventure Sonic Shuffle	Soul Calibur Spawn: Demon's Hand Sports Jam StarLancer St.Fighter III: 3rd Strike Stunt GP Tennis 2K2 Test Drive 6 Test Drive Le Mans Test Drive V-Rally

Nintendo GameCube			2001
Mario Party 6	Mario Superstar Baseball	Super Smash Bros. Melee	

Nintendo Game Boy Advance			2001
Advance Wars	Harvest Moon	Mario Kart	Pokémon: FireRed
Castlevania	Harv Moon: More Friends	Mario Pinball Land	Pokémon: LeafGreen
Dragon Ball	Incredibles	Mario Tennis	Pokémon Myst Dungeon
Dragon Ball Z: Buu's Fury	Invincible Iron Man	Metroid Fusion	Pokémon: Ruby
Dragn Ball Z: Supersonic	Jungle Book	Metroid: Zero Mission	Pokémon: Sapphire
Final Fantasy Tactics	King of Fighters EX	Muppet Pinball Mayhem	Super Mario
Final Fantasy VI	King of Fighters EX 2	Ninja Five-0	Super Mario 2
Fire Emblem	Leg of Zelda: 4 Swords	Pac-Man Pinball Advance	Super Mario 3
Fire Emb: Sacred Stones	Leg of Zelda: Minish Cap	Pinball Challenge Deluxe	Super Mario 4
Golden Sun	Mario & Luigi: Superstar	Pokémon: Emerald	Super Monkey Ball Jr.
Golden Sun: Lost Age	Mario Golf: Adv Tour	Pokémon: Emerald (hack)	Super Street Fighter II

Sony PlayStation Portable			2004
Age Of Hammer Wars	D&D Tactics	Marvel: Ult. Alliance 2	Spider-Man 2
Age of Zombies	Dungeon Siege	Metal Gear Acid	Tetris
Angry Birds	Final Fantasy Tactics	Metal Gear Solid: Ops	Top Gun
Arcade Darts	Ghost Rider	Metal Gear Solid: Peace	Trailblazer
Arcade Essentials	God of War: Sparta	Monochrome Racing	Twin Blades
Arcade Evolution	Golf Mania	Monopoly	Vector TD
Arcade Pool & Snooker	Halfbrick Blast Off	Monster Hunter Freedom	Virtua Tennis: World Tour
Blitz: Overtime	Half-Minute Hero	Pac-man Championship	VT Tennis
Blood Bowl	Let's Golf!	Pinball Dreams	X-Men Legends II
Bowling 3D	Lord of the Rings: Tactics	Pinball Fantasies	X-Men Origins: Wolverine
Crisis Core: FF VII	Marvel Nemesis	Sky Force	Young Thor
Duodecim Final Fantasy	Marvel: Ultimate Alliance	Space Invaders Extreme	Zombie Racers
Dracula			

💾 Current Image

13-Oct-2026

- latest working git commits off EmulationStation and RetroArch v1.22.2 release
  - rebuilt cores: Atari 800, Dolphin, DOSBox Pure, Flycast, Stella, VICE
  - maintain **primary** cheevos account credentials as a layered config over new keychain
  - dropped Animatics-DX because of segmentation faults; replaced with Zmax + random startup
- VICE updated **`sdl_sym.vkm`** for xvic, x64sc and x128 cores
  - commented out Left SHIFT and CAPS LOCK entries to **avoid core mechanical shift-lock conflicts with Game Focus** toggle - use Right SHIFT when needed
  - transposed meta keys: TAB as CONTROL and Left Ctrl as CBM to be like positional
  - VIC, C64, and C128 core configs changed to "**symbolic-user**" to use the changes
  - core hotkey for onscreen keyboard toggle mapped as Left ALT
- more improvements and additions to the curated content and supporting media

📓 Changelog

26-Sep-2026

- Debian 13 Trixie with latest dot release 7 (*note* Debian 11 Bullseye *reached its* LTS *end of life*)
- latest working git commits off RetroArch v1.22.2 release
  - replaced pulse with threaded audio using pipewire driver change
  - many fixes and updates to audio, desktop menu, accounts & statistics handling
- tweaks made to scripting, account handling, and user shaders
  - self-modify rclone.conf off Bookshelf token to avoid Retro token expiry failure
  - corrected overlays off retroarch changes
  - updated Animatics-DX and Hypermax-Plus-PixN themes
- EmulationStation gamelist corrections, docs
  - rebuilt missing NES (Mesen) core
- added a Visio-style tool to the DotNET apps

28-Aug-2026

- 🇺🇸 **MAGA** image added to Arcade Punks FTP server
- Debian 13 Trixie with latest dot release 6
  - **console playlists can only use glcore over vulkan driver as workaround for the RetroArch console regression off the MESA 26.2 package release - vulkan content runs off Desktop mode**
  - added DotNET apps to run Lonnie's reimagined classic titles
- latest git commits off RetroArch v1.22.2 release, with rebuilt cores
  - Dolphin emulator for GameCube optimized for reduced latency
  - Stella 8.0pre standalone and core reintroduced to replace Stella 2014 with added overlays
  - Desktop mode includes more comprehensive features and controls
- WirePlumber improved audio mixer volume support on-boot or switching from the desktop
- *added content is not part of the upgrade process - reimage as necessary*
  - GameCube: Mario Superstar Baseball
  - Playstation: MediEvil and MediEvil II

27-Jul-2026

- **new Debian 13 Trixie** with latest dot release 6: built+tested on Pi models: 4/400/5 **and new 500+**
  - reorganized console playlists menu into more curated lists by its era's distribution media
  - added Pi 500+ **keyboard lighting** theme with support to highlight the relevant keycaps used by running content (*whenever possible*)
  - added to desktop: printer support and icons to run playlists from EmulationStation or RetroArch
- latest git commits off RetroArch v1.22.2 release, with rebuilt cores
  - **`CAPSLOCK`** is now designated as game / keyboard toggle focus mode
- latest hypseus singe emulator
- dropped all Saturday cinema and movie content to move into **`Ageless`** FFmpeg playlist
- updated **`EmulationStation`** themes
  - added Retro_LUX, iconic-es and Hypermax-Plus-PixN
  - updated Animatics-DX and PlayStation-X theme

🗓️ Older

18-Jun-2025

- 🇺🇸 **MAGA** rebranded release: *My Awesome Games Archive*
- Debian 12 Bookworm with latest dot release 11: built+tested on Pi 4gb models: 4/400 and Pi 5
- latest git commits off RetroArch v1.21 release, with rebuilt cores, of note:
  - latest autoconfig controller profiles in support of the new input detection strategy
  - BlueMSX, Citra, FBNeo Light, Genesis Plus GX, MAME 2003-Plus, MAME 0.277, PCSX ReARMed, PPSSPP, Stella, VICE 3.9
- added launch Chromium web browser off the console menu using **`weston`**
- updated Animatics-DX and PlayStation-X theme
- latest Angband build

20-Jan-2025

- 🇺🇸 **MAGA** release: *Making Arcade Games Accessible*
- Debian 12 Bookworm with latest dot release 9: built+tested on Pi 4/400 and Pi 5
- latest git commits off RetroArch v1.20 release, with **62** cores rebuilt, of note
  - bsnes-jg, MAME 0.273, PCSX ReARMed, PUAE, VICE
- updated PlayStation-X theme to v.41 to use its PS4 style gamelists
- improved Cheevos account+password save support for more eligible Cheevos games
- improved upgrade option to manage more folders and to maintain symbolic links
- added a udev rule to address an 8BitDo retro mechanical keyboard issue with its 2.4ghz adapter that registers a "mouse" device along with the keyboard – RetroArch's Mouse Index selects it first even though there is no mouse/trackball physical device attached to it
- minor menu cosmetic changes and many updated emulator screen overlays + fixes

28-Dec-2024

- latest git commits off RetroArch v1.19.1 release, with cores rebuilt
  - bsnes-jg, Flycast, MAME 2003-Plus, MAME 0.272, PCSX ReARMed
- added **`Alt`** keypress to Playlists: left prompts to switch virtual terminal and right for splash image
- Pi 5 using kernel8 image reported an issue with its new default 64mb CMA for bug fixing
- loaded up Sega Dreamcast collection with **`40`** titles in EmulationStation
- locally added game manual links for EmulationStation collections for faster viewing
- corrected Hypseus for all laserdisc games and added default bezel
- refreshed Sinden light gun support
- improved udev detection of usable input event device(s): joystick, keyboard and mouse

25-Nov-2024

- three years of retro computing and gaming on Raspberry Pi: I rate it an **A-** experience
- refresh new Raspberry Pi OS using default LightDM with its Lab Wayland Compositor (labwc)
- Debian 12 Bookworm with latest dot release 8: built+tested on Pi 4/400 and Pi 5
- latest git commits off RetroArch v1.19.1 release, with cores rebuilt
  - Atari 800, Citra, FBNeo (Light), Gambatte, Genesis Plus GX, MAME 0.271, MAME 2003-Plus, PCSX ReARMed, PicoDrive, PPSSPP, Stella
  - add more game manual links whenever possible for its integrated PDF image viewer
- continue custom build out using **EmulationStation** with curated playlists (**700** titles)
  - updated themes and allow for user selection over the random pick before launching
  - added FakeXMB theme
  - show a random system hype video out of the PlayStation-X theme
- reduced console **tty**: 1 - playlists menu with 2 - 4 auto login **pi** shells, 5 - fbi splash, 7 - lxde
- Pi 5 eeprom config: `PSU_MAX_CURRENT=5000` with boot `usb_max_current_enable=1`

27-Oct-2024

- latest git commits off RetroArch v1.19.1 release, with cores rebuilt
  - blueMSX, DOSBox Pure, FBNeo (Light), MAME 0.270, MAME 2003-Plus, Mesen, Mupen64plus-Next, NeoCD, PCSX ReARMed, Snes9x
- WIP: custom build **EmulationStation** with curated playlists (**583** titles) since dropping CoinOps Legends RetroFE last year
  - starts up using a random 5-star theme out of a set of 4: Animatics-DX, Arcadeplanet, PlayStation-X with built-in Carbon
  - Arcade, Atari 2600 & 7800, Commodore VIC 20, 64, CD32, ColecoVision, Daphne (with latest **Hypseus** emulator), Nintendo NES, SNES, GameCube, Gameboy Advance, Sega SMS, Genesis, 32X, Saturn, Sony PlayStation, PSP, Vectrex
  - add link to my game manuals whenever possible for its integrated PDF image viewer
- Raspberry model detect in **autostart.sh** to prep best performance & visual tweak differences for Pi 4/400 and 5 startups, including which core to use: faster Snes9x or more accurate bsnes-jg
- new **ColecoVision** keyboard mapping option enabled: simply press 1 - 8 to start most games

28-Sep-2024

- Debian 12 Bookworm with latest dot release 7: built+tested on Pi 4/400 and Pi 5
  - added Endless Sky game to desktop toolbar
- latest git commits off RetroArch v1.19.1 release, with active cores rebuilt
  - Dolphin, FBNeo (Light), Flycast, Genesis Plus GX, MAME 0.270, MAME 2003-Plus, Opera, PPSPP, PUAE 5.3, PX68K, and VICE 3.8 (*with* **C128** 40/80 hotkey *symbolic mapped as* **F9**)
- replaced cores Beetle PCE FAST with Beetle PCE and snes9x with bsnes-jg for more emulation accuracy over speed hacks in playlists
- added ad hoc content off playlists menu
  - Sega Dreamcast NBA + NFL 2K2 to Niche games
  - Nintendo GameCube Super Mario Smash Bros Melee with Sega Dreamcast Gauntlet Legends and Hoyle Casino to Party games
  - 3DO Road Rash as a quick-pick

28-Aug-2024

- latest git commits off RetroArch v1.19.1 release, with cores rebuilt
  - FBNeo (Light), MAME 0.268, MAME 2003-Plus, gpSP
- updated config overlays with custom x,y offsets to use absolute (0,0) centering bias and suppressed those fixed offsets to use the new bias default centering values instead
- updated all Arcade - Analog content to use the MAME core
- added **Raiden DX** to Arcade
- replaced the more obscure MAME rom naming used for playing the attract videos value
- added custom MAME crosshair cursors for Crossbow, Seawolf, etc.

24-Jul-2024

- Debian 12 Bookworm with latest dot release 6: built+tested on Pi 4/400 and Pi 5
- latest git commits off RetroArch v1.19.1 release, with cores rebuilt
  - FBNeo (Light), Genesis Plus GX, MAME 0.267, MAME 2003-Plus, Mupen64+ Next, and VICE
- WIP to replace the more obscure MAME rom naming used for playing the attract videos
- added **`Marble Madness 2`**, **`Martial Masters`** to Arcade
- added **`Vanguard`** to my main quick picks
- added **`0 A.D.`** (akin to Age of Empires II) and latest Windows application installer (Wine)

21-Jun-2024

- latest git commits off RetroArch v1.19.1 release, with
  - assets and databases updated
  - added hotkey (**`K`**, **`L`**, **`N`**, **`P`**) support for running game Save States
  - cores rebuilt: FBNeo (Light), Genesis Plus GX, MAME 0.266, MAME 2003-Plus, NeoCD
  - PCSX ReARMed for PlayStation new TurboCD option enabled
- WIP to replace the more obscure MAME rom naming used for playing the attract videos
- WIP install mancave wallpapers (action, fantasy, pinups) as a desktop slideshow mode option
- Steam and Wine (WOW64 mode and box64) installed
- added **`Field Combat`** to main picks

31-May-2024

- latest git commits off RetroArch v1.19.0 release with cores rebuilt, of note
  - FBNeo (Light), MAME 2003-Plus, PUAE, and VICE
- always invoke GameMode when playing RetroArch off Desktop or Console – addresses the potential for significant input latency issues especially when using modern 4K/8K displays in spite forcing 1080p
- added some desktop strategy games
- enhanced Playlists menu **`upgrade`** option that adds rclone copy off this GDrive **`aarch64`** folder to apply updates from changes made within the past **9** months against RetroArch runtimes, configs, and supporting scripts

30-Apr-2024

- latest RPi eeprom firmware, Linux (6.6.28+rpt-rpi-v8), and Mesa (24.0.5) updated
- latest git commits off RetroArch v1.18.0 release with **66** cores rebuilt, of note
  - refreshed assets, controllers, databases, overlays and shaders
  - FBNeo (Light), MAME 0.265, PCSX ReARMed, PicoDrive and VICE for Commodore 8-bit computers
  - VecX changed to use software renderer not GLES3
- added Tandy and Texas Instruments (MAME) to Museum list, formerly Computers
- added Intellivision artwork, game bezels and manuals
- console playlists menu reorg with enhancements and reorganized quick-picks into **`.local`**
  - **`main`** menu lists the simpler 1 or 2 player arcade games
  - added subcategories: Board & Party, Pinball, RTS & Turn-based, Niche Controllers, and Homebrews
  - added a play VT240 terminal emulator script to telnet play.ddgame.us

28-Feb-2024

- Debian 12 Bookworm with latest dot release 5: built+tested on Pi 4/400 and Pi 5
  - latest Linux (6.6.18) with firmware (Feb 16)
  - Pi 5 firmware setup for 15% overclocking with boot order: SD card, NVMe drive, USB stick
- latest git commits off RetroArch v1.17.0 release with active cores rebuilt, of note
  - FBNeo (Light), MAME 0.262 and moved off Stella and onto Stella 2014
  - added Sega Naomi2 (Flycast) support
- KDE Plasma 5 updated
  - Wayland autostart scripts include changing display settings to HD + 120% scaling
  - bluetooth adapter automatically detects & powers on at desktop (or console) startup
  - desktop RetroArch apps run content out of a Wayland window properly, but note that Nintendo 64 and arcade vector games still run their best out of a console screen
  - added Chromium apps and the Pi-Apps store
- added mouse button + wheel controls to console playlists menu

26-Jan-2024

- latest Linux (6.1.74), firmware (Jan 24), and Mesa (23.2.1) updated
  - boot config enable PCI express for NVMe as gen 3
  - `sudo rpi-eeprom-config --edit` commented entries to boot off NVMe
- latest git commits off RetroArch future v1.17 with **61** cores rebuilt
  - converged non-working MAME 2003-Plus content to latest MAME 0.261 core with artwork
- added desktop menu favorites shortcut to Commodore VIC 20 with the Mega-Cart attached
- added desktop PS2 shortcut to AetherSX2 emulator with a games playlist
  Note: *the sizable content is missing until it is actually playable - works on Linux x64* Ryzen 9
- added machines and content to playlists
  - Apple IIgs (MAME) to Computers
  - Mattel Intellivision (FreeIntv) to Consoles
  - Pokémon Mini to Handhelds
  - Apple models and Sharp X68000 (PX68k) and modified Commodore models to Museum
  - TIC-80 to Big Menu

25-Dec-2023

- Debian 12 Bookworm with latest dot release 4: built+tested on Pi 4/400 and Pi 5
  - Pi 5 boots using kernel8.img and includes latest firmware
  - Mesa lib refresh – Vulkan GPU screenshots are working
- latest git commits off RetroArch base v1.16.0.3 release
  - dropped gl (OpenGL 2.0+) in favor of glcore (OpenGL 3.1+) video driver
  - udev rule and custom patch to sysled driver to enable the keyboard LEDs
  - latest git commits for DOSBox Pure and Mupen64Next-Plus cores
- desktop menu and UI improvements
  - added **F11** hotkey in desktop mode to toggle fullscreen
  - Saturday shows off desktop uses MPlayer GUI
  - flatpak integration with KMahjongg and KPatience installed
- modified Bubble Bobble quick start to have Cheevos compete or just play options
- added some game content requests
  - 40 Winks, Tomba!, and Tomba 2 (PlayStation)
  - Secrets of Monkey Island (Sega CD)

25-Nov-2023

- Debian 12 Bookworm with latest dot release 2 (built for Pi 4/400) and updates
- KDE Plasma desktop improvements with bluetooth control + gamemode enabled
- added 8BitDo Retro Mechanical Keyboard volume control in console menu
- added cc65 6502 compiler suite – I use its ca65 assembler myself
- corrected lib installs for Daphne 32-bit build
- dropped CoinOps Legends (incompatible) keeping BezelDay overlays for matching arcade content
- latest git commits off RetroArch base v1.16.0.3 release
  - added Qt 5 and Wayland support to run in a Plasma desktop window
  - built Final Burn Alpha (Light) core
  - both gl and vulkan support can run in dedicated console kms mode

24-Sep-2023

- added KDE Plasma with autologin as the default startup
- installed my personal TTF font library
- added trailers to some Saturday show selections
- replaced timeout alarm to send SIGTERM instead of SIGQUIT to retroarch

26-Aug-2023

- added audio volume adjustment operator controls to console menu playlists
- updated PlayStation Personal core for Linux 6.1 (Debian Bookworm)
- added support for PlayStation 5 DualSense controller detection with LED support
- removed 10gb of local Bookshelf manuals; remote mount cloud drive for latest access
- added Bruce Lee movies to Saturday Theater

🧑‍💻 About

`https://robert.hurst-us.net/rob/retrocomputing`



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